Requirements Specifications

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Team LumberHack

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Version 1.0

Client Signature:_____

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Introduction

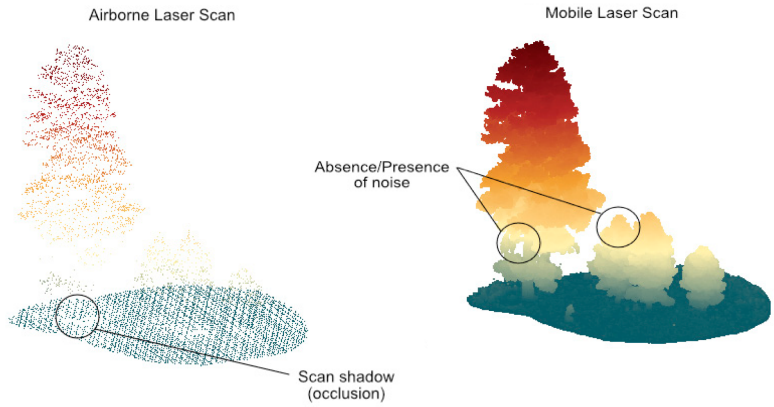
Forest ecosystem health is at the center of many large-scale environmental problems that the world faces today. Over the last century of human expansion and development, forest management policies have heavily focused on timber production and fire exclusion. The focus on production has led to high tree densities and more combustible material in forests. This has resulted in a greatly increased risk of catastrophic wildfires, droughts lasting longer and happening more frequently, as well as increased recruitment of invasive species taking over forests. With climate change advancing, forest ecosystems are facing greater threats than ever before. Scientists and forest managers work on restoration projects to try and bring forests back to a healthy state. The scale of these projects ranges from a few acres of land to many square miles of forests. The sponsor for the project is Dr. Andrew J. Sánchez-Meador. He is an Associate Professor at the School of Forestry, as well as the Executive Director at the Ecological Restoration Institute at Northern Arizona University. Dr. Meador's research focuses on using quantitative forest ecology for multi-scale forest restoration and ecology issues. By using remote sensing technologies such as light detection and ranging, or lidar, Dr. Meador and other ecologists work on applying practical interpretations of large complex data sets to model individual tree growth and forest health.

Ponderosa Pine Forest
Flagstaff, AZ

Problem Statement

Due to forests covering large spatial areas over difficult terrain, observing individual trees and capturing data can be challenging. Lidar serves as a way to collect data over these large spatial areas of forests. Lidar is a remote sensing method that uses light pulses combined with GPS data to create precise, three-dimensional information about the shape and characteristics of the trees and surrounding environment. Airborne lidar scanning (ALS) has been the traditional

method when using LiDAR for surveying, utilizing a lidar module attached to an airplane or helicopter it allows for a large area of land to be surveyed. With airborne lidar, there is more land coverage than other forms, such as mobile lidar, but the number of data points is low (around 10 points per square meter) and the forest is viewed from above. With mobile lidar scanning (MLS), data is taken from the perspective of someone walking through the forest. Mobile lidar can have a couple thousand points per square meter which results in much more data and a much denser point cloud. Current tools are built around airborne lidar and usually require programming skills to use effectively. These tools do not include graphical user interfaces (GUI's) or functions that allow for analysis of individual trees, which is important information for ecologists to have. With MLS gaining popularity and users, mobile lidar specific tools are needed. Researchers, such as Dr. Sánchez-Meador, need a tool that is built around mobile lidar data and will yield information necessary to restoration projects.



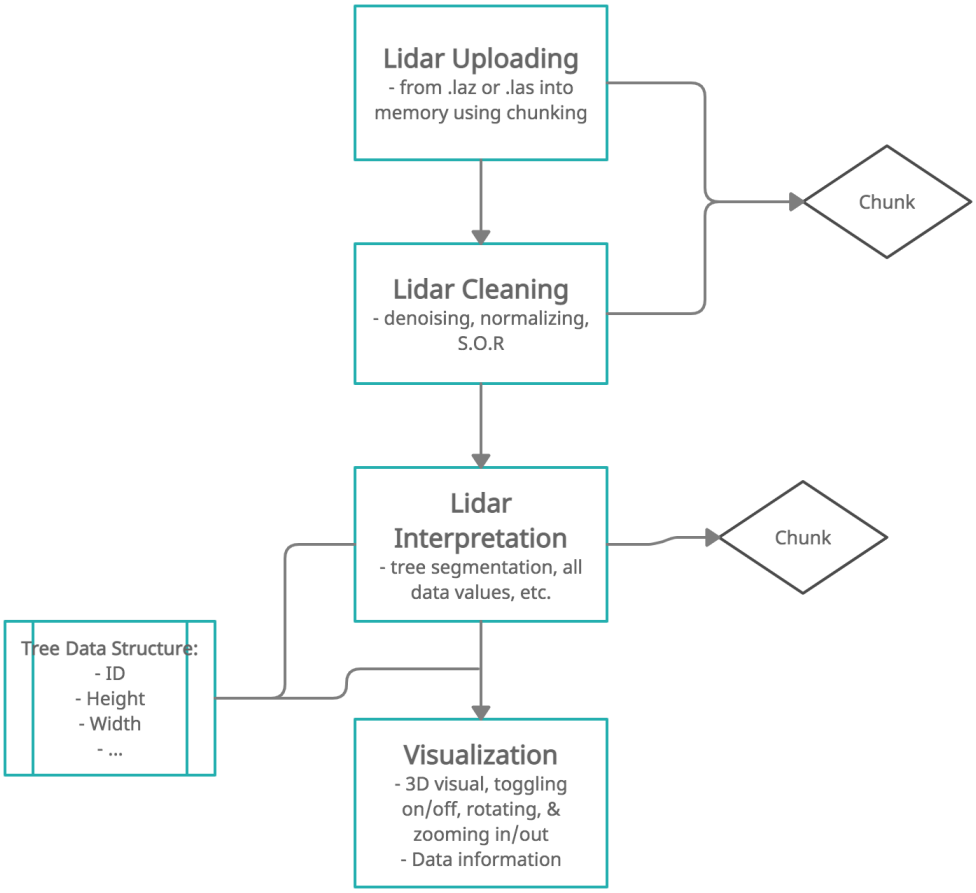
Solution Vision

This project will serve as a resource for researchers and ecologists to analyze their own MLS data. The team plans to create a web application that allows user interaction without a steep learning curve.

Specific features in the final application include:

- Ability for users to upload their own MLS data
- The data will be cleaned and normalized
- Using a GUI, the user will be able to process the data and receive useful information such as tree height, tree diameter, lowest branch height, crown size, as well as the ability to distinguish trees from smaller shrubbery
- An intuitive way to view statistics on a 3d point cloud or table

The final product will be a single portable tool that any ecologist can pick up and use. With a straightforward GUI, the team plans for users to be more focused on their research and their data than on learning a new piece of software. The project will focus on creating a mobile lidar specific application to support a growing number of MLS researchers.



Solution Vision Workflow

Project Requirements

This project will have clearly defined requirements to ensure that the team develops a tool that fits the needs of the client. The requirements will be laid out into three sections; Functional, Non-Functional, and Environmental.

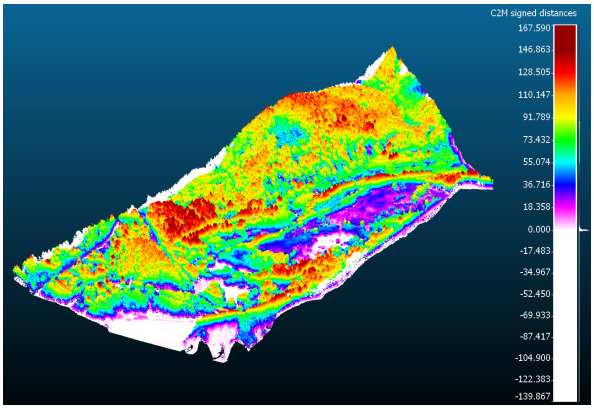
Functional Requirements

Identification of key characteristics within a point cloud

One of the key requirements necessary for the project is to be able to extract key features from point cloud data. These features of interest have to do with how groups of points are arranged and how they can be described as being more linear, planar, or spherical in orientation. Other key characteristics to be identified is the center and radius of a circle that best fits a group of points in a tree segment. Both of these methods need to be done quickly by improving current methods used in lidR and CloudCompare that do not utilize multiple processors or do not have efficient memory calculations. Once these features have been derived from the point cloud they can then be used for the creation of the 3d representation of the point cloud.

Has the ability to calculate differences between observed ecologically pertinent attributes

Another function that this application will include is having the ability to calculate differences between observed ecological pertinent attributes. These ecological attributes include height, volume, above-ground biomass, and individual tree location at two points in time. Since mobile LiDAR is capable of scanning while moving, it allows the team to get effective data collection over wide ranges. With this advantage, tree height and volume are able to be estimated using mobile LiDAR. There are preprocessing steps on the mobile LiDAR data before calculating differences in observed ecological pertinent attributes. The CloudCompare

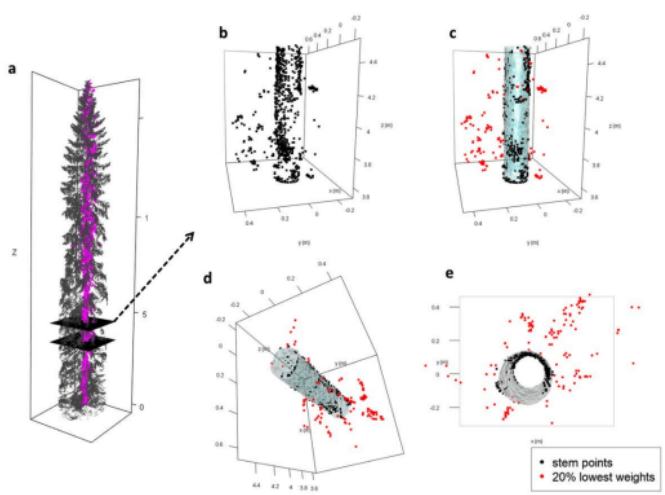


software could be used to preprocess these steps. First, the data would have to be denoised with a noise filter tool, then use CSF(Cloth Simulation Filter) filter for ground classification, and lastly, manually classify the trees and segment them into individual trees.

To calculate the tree height, the team will use the height-above-ground method. The height above ground is calculated by subtracting the height value of the highest point in the tree point cloud from the height of the ground surface. The team will use CloudCompare for the above-ground calculation using the cloud/mesh distance tool and compute the maximum height above the ground of each tree with MATLAB. The volume of the tree is computed by the stem curve. This is automatically derived from the point cloud, between the minimum height and height of the ground surface. A method to estimate above-ground biomass of large trees is by three-dimensional tree modeling of the MLS point clouds. To get accurate measurements, cylinder shape fitting is needed.

Cylinder (or mesh or multiple sphere) shape fitting with tree size at 1.37m

Incorporating the functions above, the application will also include the function of cylinder, mesh, or multiple sphere shape fitting with the tree center size at 1.37m from 1.17m to 1.57m and 0.4m to 0.5m segment. This will use RANSAC(Random Sample Consensus) for tree boles with attribute and error reporting. RANSAC extracts arbitrary random minimal sets from the point data. Even with the proportion of outliers present, it is still a reliable process. The process starts with taking a slice of a tree and separating trees. In order to get a good cylinder fit, there needs to be about 15 to 30 random points and this should be done about 10 to 20 times.



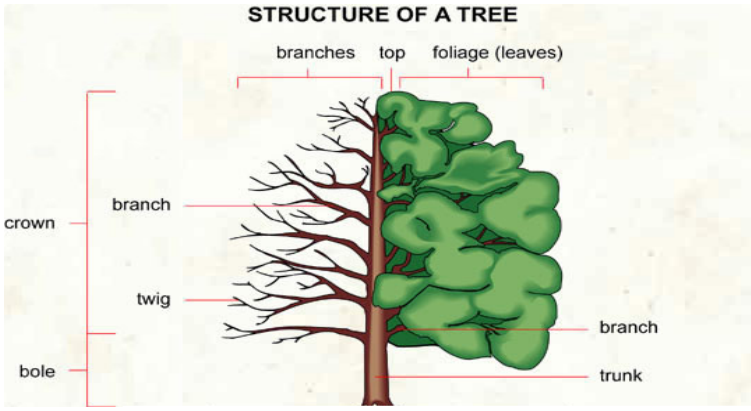
Using the Principal Component Analysis(PCA), cylinder fitting is used to estimate above-ground biomass and tree location. First, the points will outline a 1.37m sample circle. Then the tree location is defined with the projection of the circle's center. For error

reporting, about a 20% margin of error and 80% accuracy would be great for the team's software. Even if the error of what is given from a 1 inch tree compared to a 20 inch tree will be the same.

Classification of specific points

The application will include functions that allow for classification of points in the 3d point cloud. When analyzing point cloud data of a forest, a key component of forest health is the composition of trees and surrounding plant life and shrubbery. These classifications will be used for grouping of similar points, as well as breaking down individual trees into their many sections. The application will include point classifications such as the bole, branch, and crown of a tree, and others including logs that are laying horizontally on the ground.

The classification component is important for identifying composition of individual trees as well as the ability to view the point cloud with specific groupings. For example, low branches and lots of dead wood on the forest floor act as fuel for fires. Being able to classify and view just branches and dead trees on the ground would greatly improve the time it takes to clean up high fire danger materials.



Functions will be implemented in the backend of the application that identify and classify which points are which. This will be designed using algorithms that identify verticality and horizontality as well as grouping algorithms such as nearest neighbor to classify points into the groups mentioned above. The classifications will then be implemented into the GUI of the application to allow the user to select which set of points they want to view.

Registration of two, temporally distinct but overlapping, point clouds

One of the important uses of the software is to be able to monitor changing forest conditions over time. To accomplish this two point clouds need to be loaded into the program,

and then be properly aligned so that accurate comparisons can be made of the same plot of land. Due to difficulty in aligning point clouds precisely, the software will use point cloud specific tools that can fine tune the orientation of the point clouds and identify the best fit.

User interface using Shiny

_____The final application will implement the R package Shiny for its user interface. Shiny is a package that provides a web framework for building applications using the R language. The application will be GUI based and allow for easy manipulation and observation of MLS data. As a web based app, it will be portable and accessible on most machines. Functions that are implemented will require minimum programming knowledge to use and will be mostly based on GUI features such as buttons, menus, and 3d point cloud visualization.

Example Shiny Web App

The application will provide a way to allow users to upload their own MLS data. The data will then be pre-processed and cleaned using functions that are built into the lidR package that Dr. Sanchez-Meador has helped develop. Included in the GUI will be a way to allow users to modify RANSAC parameters for cylinder fitting and parameters for point classification. There will also be a way for users to view and visualize the data. This will be shown through tabular summaries (2 dimensionally) and through point cloud visualization (3 dimensionally). Structural attributes (tree height, diameter, etc) will be shown as well as the ability to color points by treeID, margin of error, and cylinder/sphere overlays.

The web app will be an all in one tool that abstracts away the technicalities of traditional lidar point analysis. The application will include tools and functions that provide a basis for ecological research projects through its Shiny based GUI.

Stretch Goals

Desirable stretch goals fall into one of two general categories: making the software more easily *usable* and making the software more easily *reusable*. On the usability front, the team envisions small "vignettes" throughout the software, textual or graphic, that explain to the user how to best use/interact with a given feature or tool, or give information about what is going on behind the scenes. These short bits of documentation need not be limited to simply explaining the user interface--they could also give insight into certain design software decisions that more advanced users, such as researchers, would find helpful and informative as they use the software. In this example, this would save the researcher from needing to go search through the official documentation for small but important pieces of information. What types of vignettes to create and where in the software UI to place them will become clearer during actual development.

On the reusability front, the team has one major stretch goal: to package the software as a CRAN package. CRAN, which stands for Comprehensive R Archive Network, is the official repository for third-party software packages developed in R. CRAN imposes a variety of criteria on software packages that wish to be officially hosted, including documentation, error handling, and testing requirements. While making the software CRAN-compliant will certainly require additional work, it will make it more visible to the scientific community, and thus more likely to be used. Furthermore, because of the strict functional requirements CRAN, aiming for acceptance into the repository will ensure a well-designed and fully functional piece of software at the end of the project. In other words, it will serve as a good metric for project success.

Non-Functional Requirements

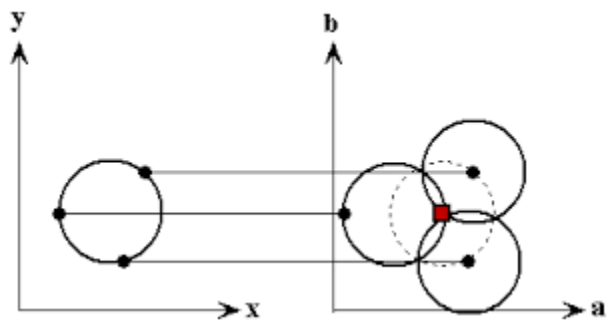
Identification of key characteristics within a point cloud

In order to identify the key characteristics in the point cloud the software will perform linear algebra calculations to determine how groups of points are oriented, as well as feature extraction techniques to identify the center and radius of a circle that best fits the points.

Using linear algebra, eigenvalues will be calculated and saved to be used in the calculation of linearity, planarity and sphericity. These values can then aid in the identification of

how these points are oriented. Once the orientation of points in a cloud have been determined they can then be used later in the classification of tree structures.

Another feature of interest is determining the best fit circle for a group of points, to do this a Hough transformation will be done. A Hough transform can be described as taking an iterative approach of an increasing radius and drawing a circle around a random sample of points using that radius. Using this collection of circles the program will identify a point that has the highest number of circle intersections and then test the fit of the candidate center and radius. Once a candidate center and radius is able to encapsulate 80% of points within a threshold distance of the circle's edge it will be used as the best fit circle for a group of points.



_____These features can be time consuming to calculate due to the linear algebra needed for solving for matrix determinants to get eigenvalues and the iterative approach used in the Hough transform to find the center and radius of a circle. Here efficient use of previously calculated eigenvalues and multithreaded computation is needed to derive the key features needed for 3d visualization.

Has the ability to calculate differences between observed ecologically pertinent attributes

_____To clearly visualize the ecologically pertinent attributes of trees, the data will go through preprocessing steps. The data has to be denoised first with the noise filter tool in CloudCompare. Then the CSF filter will allow users to see points from the ground until the tops of the trees. Lastly, the team will manually classify the trees and segment them into individual trees. After the data is pre-processed, the calculated attributes of large trees will be visualized by three-dimensional tree modeling of the MLS point clouds. To get a clear display, cylinder shape fitting using RANSAC is needed.

Cylinder (or mesh or multiple sphere) shape fitting with tree size at 1.37m

_____RANSAC requires to get about 15 to 30 random points for a good cylinder fitting and needs to do this process 10 to 20 times. It is expected that it performs with high speed to return the cylinder. The client states that minutes are fine and ideal, but seconds would be golden. For error reporting, about a 20% margin of error and 80% accuracy would be great for the team's software.

Classification of specific points

Classifying of specific points allows for breaking down the point cloud to allow for viewing and analysis of specific sections. The project should minimally have the ability to classify points as ground or bole. The client has stated that 80% accuracy for classification is a good baseline to hit, and anything above that would be great. Classifications for branches and crown of a tree would be a nice inclusion but are not a minimum requirement for the project. The classification algorithm should take no more than 1-2 minutes as the eigenvalues allowing for classification will already be calculated and ready to use.

Registration of two, temporally distinct but overlapping, point clouds

There are two options that will be provided to users in order to fulfill this requirement. First, highly reflective markers can be used as a point of reference if both scans include them and have been placed in the same location. Second, point clouds can be manually rotated by a user within the GUI by clicking and dragging the point clouds into a rough estimation of being aligned.

Once the scans have been aligned a registration tool such as the CloudCompare Iterative Closest Point(ICP) algorithm can be used to finely tune the orientation of the two point clouds. After the ICP algorithm has completed comparisons can be made to determine what changes have been made in the given plot.

User interface using Shiny

The user interface (UI) will be clear and easy to use in its capabilities. This means that only necessary functionality will be included such as the ability to clean and process data to produce meaningful results in both tabular and 3 dimensional forms. It will be easy to use as descriptions and more information on specific functions will be provided to the user. The package will be portable and on the web allowing for anyone to access and use it.

Environmental Requirements

Few environmental software requirements were imposed by the project's client. The most significant constraint imposed is that the team use the R programming language and environment. There are multiple reasons for this. First, the R environment sees widespread use in the scientific community, and is a relatively straightforward language for non-programmers to understand and write. This latter aspect is important for the project because the team is trying to make complex software accessible. Furthermore, many contemporary software tools for lidar processing are written in R, including LidR, the package from which the team plans to borrow certain functionalities. Second, the project's client is most experienced with the R programming language, which means he will be able to best give us guidance using this environment.

One further environmental restriction is the proprietary GeoSLAM lidar processing algorithm that is run on the raw lidar output from the mobile lidar machine as the very first step in the processing pipeline. This step is restrictive only insofar as it must be performed; the raw machine lidar output is uninterpretable without this processing anyways. The output from this process is in the standard .las file format.

Beyond these requirements, essentially no others were given. Because of the novelty of the project, the open-endedness of the research context, and the fact that the team will be implementing nearly an entire pipeline from raw lidar data through data visualization, there are few opportunities to be significantly restricted by environmental software. That being said, as development continues, and as yet unspecified third-party software is incorporated, other environmental requirements are likely to arise.

Potential Risks

The team anticipates three general areas of risk associated with the final project. These are: inaccurate or invalid outputs from the core processing functionalities, poor code development and/or poor documentation, and a poorly implemented user interface. Each of these is important for a distinct reason.

First, giving false or inaccurate outputs from the core processing tools would be problematic for many obvious reasons. Any downstream data reporting, including data visualizations, would be invalid, or at best misleading. Even slightly inaccurate results, or many outputs of which only a small subset are invalid, are problematic because they may degrade confidence in the software and deter its use by researchers. In short, accurate and correct calculations must be the foremost concern during development of the software.

Second, poor code development and documentation are also problematic for multiple reasons. Convoluted code or poor documentation would deter collaborative development from other researchers, and would likely frustrate first-time users who can not make sense of the documentation. Further, code maintenance would become difficult, leading to the risk of disuse and eventual project obsolescence.

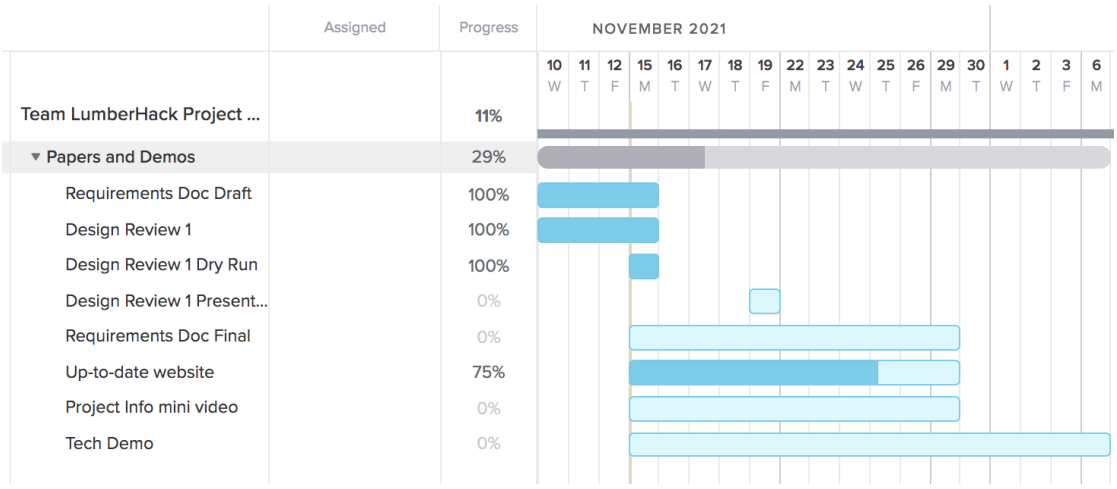
The third risk is the development of a user interface that is inadequate in one or more ways. Because the UI will be the face of the software package, it is crucial that it be intuitive, able to control all necessary aspects of the underlying software, and be non-buggy. An unintuitive or buggy UI will frustrate first-time and non-specialist users, possibly causing them to give up on using the software altogether. A UI that is not capable of performing all desired functional requirements will render those underlying software features unusable and thus wasted.

Project Plan

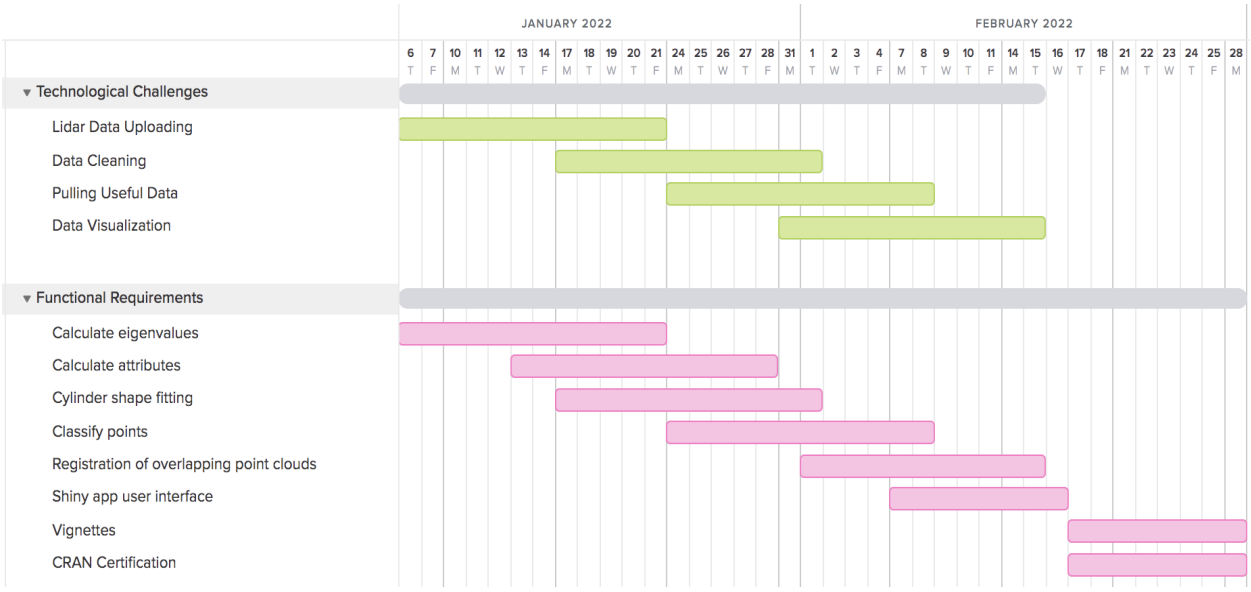
The purpose of this project is to serve as a resource for researchers and ecologists to analyze their own MLS data. So, the team's goal is to create a web application that allows user interaction without a steep learning curve. As of right now, eight milestones have been identified. The major points of the team's plan is to calculate the decomposed eigen and hough transformation values, calculate differences between observed ecologically pertinent attributes, use RANSAC for cylinder shape fitting with tree size at 1.37m, classify points, register two

temporally distinct but overlapping point clouds, and using Shiny for user interface. The team's stretch goals are to incorporate vignettes to explain the functionality of the package and the Shiny appPackage uploaded and passing all CRAN checks so it can be an official "CRAN" package.

In order to complete the eight milestones of this project, each will be laid out for the months to come. Starting now, the team will continue to work on the tech demo and project info mini-video and have it completed by the end of the semester. Below will first show the Gantt chart of the team's plan this semester.



Once the team heads into the spring semester, there will be more specific major and minor milestones added onto the project plan. Below is a rough breakdown of the team's plan next semester, presented in a Gantt chart.



Conclusion

Through the experience gathered by Dr. Sánchez-Meador as the executive director of the Ecological Restoration Institute, it has been found that there is a lack of efficiency in the current methods used to monitor forest health. Traditional methods typically involve groups of three individuals manually measuring trees, recording data, and then later uploading collected data to a computer for processing. This method is time consuming, labor intensive, does not provide a full picture of the forest structure, and falls short on the interpretability of the results. More modern approaches that use LiDAR are implemented with the idea of airborne LiDAR monitoring, providing a top down view of the forest and not a ground level view. Require technical knowledge that can be challenging for non-technical individuals, or the software used does not utilize hardware efficiently in the processing of data.

This project intends to solve these problems by providing a more efficient method of forest monitoring. First it will require less people for data collection by using mobile LiDAR to create a point cloud of the forest. From the point cloud the project will then provide an easy to use interface via a Shiny App for the upload, cleaning, interpretation, and visualization of lidar data. From the functionality of the app users will be provided with a tool that can effectively derive and display pertinent ecological data, as well as see ground level views of the forest structure all within an easy navigable web app.

So far requirements have been gathered over meetings with the sponsor and technological feasibility has been conducted on possible implementation methods. With a clear understanding of the project needs and possible tools to address them, the team looks forward to implementing a solution to these challenges.

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